

The Intelligent Complaint Classification Using Localized Transformer Architecture project addresses telecommunications service delays in Nigeria by automating the routing of localized customer complaints.

Purpose of the Project

- The project utilizes a multi-class text classification system to categorize natural language telecommunications complaints.
- It processes inputs containing Nigerian English, Nigerian Pidgin, local expressions, and telecom abbreviations, mapping them to core domains like Mobile Data, Network Quality, and Billing & Fraud.
- By identifying actionable intents such as frequent call drops, abnormal data depletion, or unexplained credit deductions, the model routes issues directly to the correct resolution departments.

Positive Impacts

- The model accelerates support ticket resolution and eliminates systemic service delays by preventing complaints from being routed to the wrong departments.
- It directly aids in rebuilding trust for over 184 million telecom users in Nigeria.
- Communities gain fair access to essential digital infrastructure support and experience reduced financial worry related to unexplained billing discrepancies.

Potential Risks and Harms

- Because the training data was harvested entirely from the internet, it predominantly reflects the experiences of smartphone-literate city dwellers.
- The system risks excluding rural customers with poor internet access and individuals who struggle to read or write.
- Demographics that communicate strictly in pure indigenous languages, or those who prefer to call customer care directly rather than typing online, are currently underrepresented.

Values

- The project is grounded in the core principles of fairness, accessibility, privacy, reliability, and accountability.
- By actively recognizing local slang and Nigerian Pidgin, the system ensures that non-standard English speakers receive an equal quality of automated customer service regardless of their formal English proficiency.

Mitigation Strategies

- To uphold stringent privacy standards, all publicly sourced data was stripped of Personally Identifiable Information (PII), including names, phone numbers, subscriber account balances, and banking details.
- To counteract representation disparities, targeted synthetic data generation was integrated to capture low-resource Nigerian Pidgin syntax and regional phrasing variations.
- Algorithmic bias and performance drift will be actively prevented through continuous open audits, iterative feedback mechanisms, and regular benchmark evaluations on localized holdout sets.

Limitations and Uncertainties

- The dataset currently relies on AI-generated labeling, which necessitates ongoing consistency checks by human reviewers fluent in local dialects.
- To fully reflect the diversity of the community and address current coverage gaps, future iterations of the dataset must successfully incorporate text transcripts from offline call centers.